

1. Administrative & Study Identification

Full Study Title: A Phase 3 Study of Pembrolizumab in Combination With Carboplatin/Taxane (Paclitaxel or Nab-paclitaxel) Followed by Pembrolizumab With or Without Maintenance Olaparib in the First-line Treatment of Metastatic Squamous Non-small Cell Lung Cancer (NSCLC) **Short Title/Acronym:** A Study of Pembrolizumab (MK-3475) With or Without Maintenance Olaparib in First-line Metastatic Squamous Non-small Cell Lung Cancer (NSCLC, MK-7339-008/KEYLYNK-008) **Protocol Number:** MK-7339-008 **NCT Number:** NCT03976362 **Posting ID:** 324210250010471026 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Pembrolizumab (Keytruda®), Olaparib (Lynparza®), Carboplatin (Paraplatin) and Taxane (Paclitaxel or Nab-paclitaxel) **Phase of Development:** Phase 3 **Medical Monitor:** Clinical Operations Lead, Merck Research Laboratories

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare progression-free survival (PFS) per RECIST 1.1 by blinded independent clinical review (BICR) and overall survival (OS) of pembrolizumab plus maintenance olaparib versus pembrolizumab plus maintenance placebo in patients with metastatic squamous NSCLC. *Note: As of Amendment 07, there will be no further analyses for OS and patient-reported outcome assessments.* **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Duration of Response (DOR), safety profile (including number of participants with one or more Adverse Events [AEs] and those who discontinued intervention due to an AE), and health-related quality of life assessments (EORTC QLQ-C30).

2.2 Background & Rationale While significant advancements have been made in lung cancer treatment, metastatic squamous NSCLC remains a challenge. PARP inhibitors, such as olaparib, can upregulate PD-L1 expression and promote immune-mediated responses. Adding a PARP inhibitor to first-line anti-PD-1-based therapies may improve and prolong the efficacy of immunotherapy in this patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Triple-blind per sponsor design) **Randomization:** 1:1, Stratified by ECOG

Performance Status, PD-L1 TPS expression, and response to induction treatment

3.2 Planned Enrollment & Duration Target **\$N\$:** 851 participants
Number of Sites: Global multicenter **Estimated Study Start:** 2019
Estimated Primary Completion: December 2023 (Study discontinued early for futility based on the recommendation of an independent data monitoring committee).

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Previously untreated stage IV non-small cell lung cancer (NSCLC) with confirmed squamous histology. 3. Measurable disease per RECIST v1.1. 4. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1 assessed within 7 days prior to administration. 5. Achieved Complete Response (CR), Partial Response (PR), or Stable Disease (SD) after 4 cycles of induction therapy (pembrolizumab + chemotherapy) to qualify for randomization into the maintenance phase. 6. Provided archival tumor tissue sample or newly obtained core/incisional biopsy.

4.2 Exclusion Criteria 1. Prior systemic treatment for advanced/metastatic NSCLC. 2. Known additional malignancy that is progressing or has progressed within the past 3 years requiring active treatment. 3. History of active autoimmune disease that has required systemic treatment in the past 2 years. 4. History of non-infectious pneumonitis/interstitial lung disease (ILD) that required steroids, or current pneumonitis. 5. Known hypersensitivity to any components or excipients of carboplatin, paclitaxel, nab-paclitaxel, or olaparib; or severe hypersensitivity ($\geq$ Grade 3) to pembrolizumab.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Induction Phase:** Pembrolizumab 200 mg IV Q3W + Carboplatin (Paraplatin) (AUC 6) + Paclitaxel 200 mg/m² or Nab-paclitaxel 100 mg/m² for 4 cycles. * **Maintenance Phase:** Pembrolizumab 200 mg IV Q3W (up to 31 cycles) + Olaparib 300 mg orally twice-daily (BID) or matching placebo. **Route of Administration:** Intravenous (Pembrolizumab, Chemotherapy) and Oral (Olaparib/Placebo). **Duration of Treatment:** Induction for 4 cycles (~12 weeks), followed by Maintenance for up to 31 cycles (approx. 2 years) or until centrally verified progressive disease (PD), intolerable toxicity, or physician decision. *Note: As of Amendment 07, participants actively taking placebo will discontinue taking the placebo intervention and continue in the study.*

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-induced side effects.

Prohibited: Concurrent experimental anti-cancer therapies, systemic corticosteroids exceeding standard physiological doses, live vaccines, and therapies strictly conflicting with PARP inhibitor use.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor Imaging, PD-L1 Testing, Medical History, Baseline Labs
Induction	Cycles 1-4	Pembrolizumab + Chemo Induction (Q3W), Safety Labs, Efficacy Assessment
Maintenance Baseline	Post-Cycle 4	Randomization, First Dose of Olaparib/Placebo
Interim	Cycles 5-35	Safety Labs, Maintenance Therapy (Q3W), Tumor Imaging (RECIST 1.1)
End of Study	At Progression	Treatment Discontinuation, 30-Day Safety Follow-up, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) and Overall Survival (OS) measured from the time of randomization. **Measurement Method:** Blinded Independent Central Review (BICR) evaluating tumor response via RECIST v1.1. *(Per Amendment 07, no further analyses for OS).*

7.2 Secondary & Exploratory Endpoints * Number of Participants With One or More Adverse Events (AEs) * Number of Participants Who Discontinued Study Intervention Due to an AE * Change From Baseline in European Organization for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire-Core 30 (QLQ-C30) Global Health Status (GHS)/ Quality of Life (QoL) (Items 29 and 30) Combined Scale Score *(Per Amendment 07, no further analyses for patient-reported outcome assessments).*

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection by investigators at each cycle visit, continuing through 30 days post-last dose. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) oversaw the study. Following Interim Analysis 3, the IDMC recommended halting the study for futility as it did not demonstrate significant OS or PFS improvements.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for all randomized participants. **Sample Size Justification:** Powered to detect statistical superiority of pembrolizumab + olaparib for dual primary endpoints (PFS and OS) against predefined alpha boundaries (e.g., 1-sided $P = 0.003202$ for PFS and $P = 0.014647$ for OS). **Handling of Missing Data:** Standard censoring for time-to-event outcomes per the protocol SAP guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits by the sponsor. **Audit Trail:** eCRF locking, tracking of safety alerts, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-7339-008 **Registry Number:** NCT03976362 **Posting ID:** 324210250010471026 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor/Lead Organization:** Merck **Study Title:** KEYLYNK-008 **Official Regulatory Title:** A Phase 3 Study of Pembrolizumab in Combination With Carboplatin/Taxane (Paclitaxel or Nab-paclitaxel) Followed by Pembrolizumab With or Without Maintenance Olaparib in the First-line Treatment of Metastatic Squamous Non-small Cell Lung Cancer (NSCLC)

12. Clinical Framework (NSCLC Specific)

Disease Areas: Carcinoma, Squamous Cell, Non-small-cell Lung

Preferred UMLS Name: Metastatic Squamous Cell Lung Carcinoma

Diagnosis Threshold: Stage IV, Confirmed Squamous Histology

Medical Methodology: First-line Induction Therapy followed by

Maintenance Therapy **Optimization Target:** Disease Control (PFS and OS improvement)

13. Study Procedures & Methodology

Management Pathway: Pembrolizumab + Chemotherapy Induction followed by PARP Inhibitor Maintenance **Education Protocol (HCP):** Protocol-specific AE management (immune-mediated and PARP-related toxicities) **Education Protocol (Patient):** Informed consent and oral medication compliance (olaparib) **Quality Audit Mechanism:** Independent Data Monitoring Committee (IDMC) and Central Imaging Review (BICR)

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 5 years (including survival follow-up) **Onsite Visit Cadence:** Every 3 weeks (Q3W) during the treatment phase **Remote Monitoring Frequency:** Periodic follow-ups post-progression for survival tracking **Checklist Review Interval:** Prior to each treatment cycle (Q3W) for safety labs and dosing clearance

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				